

Mathematical Frameworks — Module 07:

Communication — Study Questions

1. How is communication formalized as coupled inference?
2. What happens mathematically when Agent A speaks and Agent B listens? ($o_B = f(a_A)$)
3. What is coupled free energy? How does it differ from single-agent free energy?
4. What is generalized synchrony? How is it measured (mutual information)?
5. What does it mean for $I(s_A; s_B) > 0$?
6. How does the coupling term in F_{total} ensure belief convergence?
7. Formalize language: What do words, grammar, and pragmatics correspond to in the POMDP?
8. How does a sentence compress a generative model?
9. What are cultural norms in mathematical terms? (shared D vectors/priors)
10. How do shared priors reduce collective prediction error?
11. What happens when agents have very different priors? (miscommunication)
12. How does the multi-agent framework explain echo chambers?
13. What is the role of precision in social communication?
14. How does the framework handle deception? (intentionally generating misleading observations)
15. Can generalized synchrony be partial? What does partial synchrony look like?
16. How does turn-taking in conversation implement coupled free energy minimization?
17. What mathematical property makes human language so efficient for transmitting generative models?
18. How does this math framework differ from Shannon's information theory of communication?
19. Can Active Inference explain the evolution of language?
20. Connect this mathematical account to the cognitive science (Unit 1) and neuroscience (Unit 2) accounts of communication.